Ayush Vardhan Jha

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Education

Manipal Institute of Technology — **Manipal, India**
B.Tech in Computer Science & Engineering (AI & ML Specialization) — *Aug 2021 – Nov 2025*

Experience

AI/ML Engineering Intern — **May 2025 – Present**
TurtleLogix LLP (Client: Law Firm) — *New Delhi*

- Built production email classification system achieving 88% accuracy using Random Forest with TF-IDF vectorization on 10,000+ patent office emails across 300+ legal activity categories
- Addressed severe class imbalance (40–73% dominant classes, minority classes with <10 samples) through threshold-based filtering, controlled oversampling, and multi-model evaluation (Naive Bayes, Logistic Regression, Random Forest)
- Engineered temporal labeling pipeline from Azure PostgreSQL by correlating emails with activity timelines using 7–180 day proximity windows, improving ground-truth label accuracy
- Deployed FastAPI-based classification service with Docker containerization, batch processing endpoints, and secure PostgreSQL integration via SSH tunneling
- Developing LLM-based schema extraction for automated FER drafting using Qwen2.5-72B-Instruct with structured prompt engineering to extract issues raised, prior art citations, and deadline information

Projects

Adverse Drug Event Detection | *Healthcare NLP, Pharmacovigilance* | — **Aug – Sep 2025**

- Fine-tuned BioClinicalBERT on 23,516 medical sentences with dual-pipeline (NER + classifier) achieving **92.4% F1, 98.9% AUC-ROC** on classification and **70.8% entity F1** on drug-symptom extraction
- Handled rare event imbalance (15% positive) via weighted loss and stratified sampling; deployed Gradio demo with SHAP explainability
- Enabled real-time pharmacovigilance validation with entity highlighting and confidence scoring for clinical review workflows
- **Tech:** BioClinicalBERT, HuggingFace Transformers, spaCy, PyTorch, SHAP, Gradio

Clinical Trial Outcome Prediction | *Explainable AI, Predictive Analytics* | — **Sep – Oct 2025**

- Built multi-task XGBoost predicting treatment response (**83.2% AUC, 76% F1**) and dropout risk (**81.1% AUC, 65% F1**) on 10k Type-2 diabetes patients with 22 engineered features
- Discovered via SHAP that distance 35km+ increases dropout 12% for elderly patients; deployed interactive dashboard with targeted intervention recommendations
- Enabled trial coordinators to provide actionable interventions including transportation assistance and medication management support
- **Tech:** XGBoost, SHAP, scikit-learn, Pandas, Gradio, class-weighted loss

Medical Report Generation | *Multimodal Deep Learning* | — **Sep – Nov 2025**

- Developed ResNet-50 + GPT-2 architecture with learned projection fusing 2048-dim visual and text embeddings for automated radiology report generation, achieving **val loss 0.125** after 3 epochs
- Implemented attention mechanism enabling focus on relevant anatomical regions while generating clinical descriptions with beam search decoding
- Deployed production-ready Gradio interface with confidence estimation and fallback templates for robust clinical scenarios
- **Tech:** PyTorch, ResNet-50, GPT-2, Transformers, torchvision, Gradio, FP16 training

Publications

Soft Actor-Critic in High-Dimensional Continuous Control — IEEE 2025 | **CNN-Based Potato Disease Detection** — Springer SCI 2024

Technical Skills

Languages: Python, SQL, Java, JavaScript
ML/DL: PyTorch, TensorFlow, HuggingFace, XGBoost, scikit-learn, spaCy, SHAP
Models: BioClinicalBERT, ResNet-50, GPT-2, Random Forest, Explainable AI
Tools: Docker, FastAPI, Git, Gradio, Azure PostgreSQL, Weights & Biases